Comprehensive PDE Numerical Methods Comparison for Option Pricing

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Motivation & Problem

Motivation

Financial markets show characteristics such as volatility and sudden jumps that simpler option-pricing models cannot capture well. There lies the benefit of a framework that handles more realistic dynamics.

Problem

Different option-pricing models stem from diverse PDE/PIDE and require different numerical methods. The challenge was to build a framework that implements, compares, and validates them using real market data.

Conceptual Models

Black-Scholes Model

Assumes the underlying asset *(S_t)* follows geometric Brownian motion *(dW_t)* with constant volatility *(σ)*, leading to a one-dimensional parabolic PDE for the option price *(V)*. Other important inputs worth mentioning are the risk-free rate *(r)* and the expected rate of return *(μ)*.

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

$$\frac{dV}{dt} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

This serves as the baseline model due to its simplicity and its convenient closed –form solution for certain options: ideal for validating the numerical methods.

Heston Model

Extends Black-Scholes by allowing volatility *(σ)* to evolve randomly over time *(t)*. This introduces a second state variable: instantaneous variance (ν_t). It results in a two-dimensional PDE with a mixed derivative term. Other important inputs are the speed of mean-reversion (κ), the long-run variance (θ), and the volatility of the variance (ξ):

$$\left.\begin{aligned} dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t)dt + \xi\sqrt{v_t} dW_t^v \end{aligned}\right\} \quad dW_t^S dW_t^v = \rho dt$$

This makes the problem more realistic for capturing the randomness of volatility, but also much more computationally challenging.

Merton Jump-Diffusion Model

Adds lognormal discontinuous jumps *(J)* to the asset-price with a certain intensity (λ) using the price before the jump (S_{t-}). The pricing equation becomes a partial integro-differential equation (PIDE). Other important inputs are the average percent change caused by a jump (m) and the increment of a Poisson process (dJ_t) depicting the number of jumps in a small-time interval.

$$dS_t = (r - \lambda m)S_{t-}dt + \sigma S_{t-}dW_t + S_{t-}dJ_t$$

$$J = e^Y \qquad m = E[J-1] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1 \qquad Y \sim N(\mu_J, \sigma_J^2)$$

The model represents sudden market moves realistically, but it also introduces a nonlocal integral term that requires a different numerical treatment.

Numerical Methods

Numerical Methods for Black-Scholes

The PDE was solved with three one-dimensional finite-difference schemes: **explicit**, **implicit**, and **Crank-Nicolson**.

The basis of the schemes depends on how the spatial-time grid is defined from the Black-Scholes PDE. Due to the nature of option-pricing time marches backward as we are trying to price an option in the present from a future payoff.

$$\frac{dV}{d\tau} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

$$\tau = T - t \qquad S_i = i\Delta S \qquad \tau_n = n\Delta\tau \qquad V_i^n \approx V(S_i, \tau_n)$$

Numerical Methods for Black-Scholes

The derivative approximations used in the PDE are described as:

1. Forward time differencing: $\frac{\delta V}{\delta \tau} \approx \frac{V_i^{n+1} - V_i^n}{\Delta \tau}$

2. Central differencing for first spatial derivative: $\frac{\delta V}{\delta S} \approx \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta S}$

3. Central differencing for second spatial derivative: $\frac{\delta^2 V}{\delta S^2} \approx \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{\Delta S^2}$

To avoid overly long formulae:

$$a_i = \frac{1}{2}\sigma^2 i^2 - \frac{1}{2} r i \qquad b_i = -\sigma^2 i^2 - r \qquad c_i = \frac{1}{2}\sigma^2 i^2 + \frac{1}{2} r i$$

Numerical Methods for Black-Scholes

After algebraic manipulation the schemes can be expressed as:

1. Explicit method: $V_i^{n+1} = V_i^n + \Delta\tau(a_i V_{i-1}^n + b_i V_i^n + c_i V_{i+1}^n)$

2. Implicit method: $V_i^n = V_i^{n+1} - \Delta\tau(a_i V_{i-1}^{n+1} + b_i V_i^{n+1} + c_i V_{i+1}^{n+1})$

3. Crank-Nicolson: $\frac{V_i^{n+1} - V_i^n}{\Delta\tau} = \theta(a_i V_{i-1}^{n+1} + b_i V_i^{n+1} + c_i V_{i+1}^{n+1}) + (1-\theta)(a_i V_{i-1}^n + b_i V_i^n + c_i V_{i+1}^n)$

The spatial discretization leads to a tridiagonal linear systems which can be solved with the Thomas algorithm, an efficient form of Gaussian elimination.

Numerical Methods for Black-Scholes

Scheme	***Time Accuracy***	***Space Accuracy***	***Stability***
Explicit	1st order	2nd order	Conditional
Implicit	1st order	2nd order	Unconditional
Crank-Nicolson	2nd order	2nd order	Unconditional

Numerical Methods for Heston

From the earlier differential equations, the Black-Scholes PDE transforms into:

$$\frac{\delta V}{\delta t} + \frac{1}{2} \upsilon S^2 \frac{\delta^2 V}{\delta S^2} + \rho \xi \upsilon S \frac{\delta^2 V}{\delta S \delta \upsilon} + \frac{1}{2} \xi^2 \upsilon \frac{\delta^2 V}{\delta \upsilon^2} + rS \frac{\delta V}{\delta S} + \kappa(\theta - \upsilon) \frac{\delta V}{\delta \upsilon} - rV = 0$$

The grid notation differs from Black-Sholes by a term:

$$\tau = T - t \qquad \upsilon_j = j\Delta\upsilon \qquad S_i = i\Delta S \qquad \tau_n = n\Delta\tau \qquad V_i^n \approx V(S_i, \tau_n, \upsilon_j)$$

The ADI *(Alternating Direction Implicit)* stems from:

$$\frac{d\boldsymbol{V}}{d\tau} = A\boldsymbol{V} \qquad A = A_0 + A_1 + A_2$$

$$A_0 V = \rho \xi \upsilon S \frac{\delta^2 V}{\delta S \delta \upsilon} \qquad A_1 V = \frac{1}{2} \upsilon S^2 \frac{\delta^2 V}{\delta S^2} + rS \frac{\delta V}{\delta S} \qquad A_2 V = \frac{1}{2} \xi^2 \upsilon \frac{\delta^2 V}{\delta \upsilon^2} \kappa(\theta - \upsilon) \frac{\delta V}{\delta \upsilon} - rV$$

Numerical Methods for Heston

Instead of solving a big 2D system:

$$(I - \Delta\tau A)\boldsymbol{V}^{n+1} = \boldsymbol{V}^{n}$$

ADI solves a sequence of smaller one-dimensional systems:

$$(I - \Delta\tau A_1)Y_1 = known\ RHS$$

$$(I - \Delta\tau A_2)Y_2 = known\ RHS$$

Y_1 and Y_2 represent the solution after the implicit sweep in the S and ν direction. They are intermediate solutions.

In ADI only A_1 and A_2 are solved implicitly, A_0 explicitly or through correction stages.

Numerical Methods for Heston

Scheme	***Time Accuracy***	***Spatial Accuracy***	***Main Idea***
Douglas	1st order	2nd order in S, ν	Basic ADI splitting
Craig-Sneyd	2nd order	2nd order in S, ν	Corrects mixed derivative
Modified Craig-Sneyd	2nd order	2nd order in S, ν	Improved mixed-term stability
Hundsdorfer-Verwer	2nd order	2nd order in S, ν	Predictor-Corrector ADI

Numerical Methods for Jump-Diffusion

The risk-neutral PIDE transformed from Black-Scholes is:

$$\frac{dV}{dt} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - \lambda m) S \frac{\partial V}{\partial S} - (r + \lambda) V + \lambda E[V(SJ, t)] = 0$$

The jump term (non-local integral term) derives from the expectation as:

$$\lambda \int_{-\infty}^{\infty} V(Se^y, t) f_Y(y) dy$$

Black-Scholes uses nearby S-grid values (V_{i+1}, V_i, V_{i-1}). With jumps, a log-jump y shifts S_i to $S_i e^y$, so the option value depends on $V(S_i e^y, t)$, creating nonlocal grid dependence.

Numerical Methods for Jump-Diffusion

The semi-discrete problem is written as:

$$\frac{d\boldsymbol{V}}{d\tau} = A\boldsymbol{V} + J\boldsymbol{V}$$

$$AV = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - \lambda m) S \frac{\partial V}{\partial S} - (r + \lambda) V \qquad JV = \lambda \int_{-\infty}^{\infty} V(Se^y, t) f_Y(y) dy$$

IMEX (Implicit-Explicit) treats the local PDE part implicitly and the jump integral part explicitly:

$$(I - \Delta\tau A)\boldsymbol{V}^{n+1} = V^n + \Delta\tau J\boldsymbol{V}^n$$

- Implicit: A$\boldsymbol{V}^{n+1}$
- Explicit: J$\boldsymbol{V}^{n}$

Numerical Methods for Jump-Diffusion

If both parts were implicit:

$$(I - \Delta\tau(A + J))\boldsymbol{V}^{n+1} = \boldsymbol{V}^{n}$$

But J is nonlocal, so the matrix can become dense (containing values other than zero and complicating the tridiagonal straightforward solution).

Scheme	Time Accuracy	Spatial Accuracy	Stability
IMEX Euler	1st order	2nd order	Stable for local PDE

Monte Carlo Methods

Purpose: Independent validation tool for our PDE/PIDE solvers. The idea is to simulate the paths using each model's behavioral definition of the underlying asset and discounting the terminal payoff to find the fair price of the option.

1. ***Black-Scholes:*** simulate geometric Brownian motion paths with log-normal update.
2. ***Heston:*** simulate coupled paths for asset price and variance using correlated Brownian shocks.
3. ***Merton Jump-Diffusion:*** simulate diffusion and Poisson-driven jumps so each path includes both continuous shocks and random jump sizes.

Variance reduction: antithetic sampling (mirrors randomness on paths).

Outputs: price estimate, standard error, and 95% confidence interval.

Calibration & Data

Data

Source: We used OptionMetrics data from WRDS.

Processing: Uses NumPy and pandas to merge the raw data tables, compute meaningful quantities. Filtered the sample based on liquidity and quality constraints.

Main columns needed:

- *Contract information:* date, exdate, cp_flag, strike_price
- *Market inputs:* spot_price, rate, impl_volatility
- *Market price:* mid_price
- *Validation metrics:* delta, gamma, theta, vega

Note: Only European options considered for simplicity given a constant PDE.

Calibration

Heston and Merton Jump-Diffusion have extra model parameters that are not directly observed. These need to be estimated from the market option prices.

Heston calibration: We calibrated the five Heston parameters (κ, θ, ξ, ρ, v_0) by minimizing the difference between model prices and market prices using *nonlinear least squares* through *Levenberg–Marquardt / least-squares optimization*, following the calibration ideas from one of our sources: *Cui et al.*

Merton calibration: For the Merton jump-diffusion model, we used a *grid search* over candidate values for the different parameters (λ, μ_J, σ_J^2). The parameter set with the lowest pricing error on the sample was selected.

Calibration outputs: The calibration step produces saved parameter files which are then reused in the pricing, Greeks, and Monte Carlo scripts for both models.

Calibration Heston

Ticker	κ	θ	ξ	ρ	v_0	Feller?	CF MAE	PDE MAE
SPX	2.186	0.155	0.824	−0.902	0.030	Marginal	22.358	42.379
RUT	14.989	0.001	0.173	−0.003	0.137	Satisfied	16.031	14.861
TSLA	15.000	0.500	1.000	−0.059	0.285	Satisfied	4.021	4.706
PLTR	14.845	0.500	0.010	≈ 0	0.500	Satisfied	7.612	7.393

The Feller condition is defined as: $2\kappa\theta > \xi^2$

It guarantees strict positivity of the variance process ν_t.

When this condition is marginal, ν_t can be zero and the full truncation scheme introduces a discretization bias at each such event. This explains the noticeable difference in CF MAE and PDE MAE.

Calibration Merton Jump-Diffusion

Ticker	λ	μ_J	σ_J	MAE	RMSE
SPX	0.000	0.000	0.010	3.035	4.220
RUT	0.010	0.000	0.010	1.399	1.973
TSLA	0.010	−0.020	0.050	0.329	0.472
PLTR	0.000	0.000	0.010	0.172	0.216

The SPX and PLTR calibrations selected λ = 0, meaning the grid search found no benefit from adding jumps to the diffusion for those underlyings on this sample.

The TSLA negative mean jump is consistent with the well-known left-skew in single-stock implied volatility surfaces.

The narrow spread of MAE values across the PLTR grid reflects that options in the calibration sample were short-dated and relatively inexpensive.

Results

Results Black-Scholes

Ticker	Scheme	Contracts	MAE	RMSE
SPX	Explicit	75	1.883	2.838
SPX	Implicit	75	1.896	2.837
SPX	Crank-Nicolson	75	1.885	2.838
SPX	Monte Carlo	75	0.746	0.989
RUT	Explicit	75	1.119	1.491
RUT	Implicit	75	1.122	1.492
RUT	Crank-Nicolson	75	1.120	1.491
RUT	Monte Carlo	75	0.826	1.239
TSLA	Explicit	73	0.244	0.303
TSLA	Implicit	75	0.266	0.345
TSLA	Crank-Nicolson	75	0.265	0.344
TSLA	Monte Carlo	75	0.148	0.183
PLTR	Explicit	75	0.110	0.133
PLTR	Implicit	75	0.110	0.134
PLTR	Crank-Nicolson	75	0.110	0.133
PLTR	Monte Carlo	75	0.076	0.084

The three Black-Scholes PDE schemes produced nearly identical errors, while Monte Carlo consistently achieved lower MAE and RMSE across all underlyings.

Results Heston

Ticker	Scheme	Contracts	MAE	RMSE
SPX	Douglas	75	20.617	25.862
SPX	Craig-Sneyd	75	20.625	25.891
SPX	Modified Craig-Sneyd	75	21.306	26.496
SPX	Hundsdorfer-Verwer	75	19.463	25.167
SPX	Monte Carlo	75	24.029	30.293
RUT	Douglas	75	7.363	9.305
RUT	Craig-Sneyd	75	7.363	9.305
RUT	Modified Craig-Sneyd	75	7.430	9.365
RUT	Hundsdorfer-Verwer	75	7.248	9.220
RUT	Monte Carlo	75	7.490	9.408
TSLA	Douglas	75	0.860	0.982
TSLA	Craig-Sneyd	75	0.860	0.981
TSLA	Modified Craig-Sneyd	75	0.859	0.980
TSLA	Hundsdorfer-Verwer	75	0.862	0.984
TSLA	Monte Carlo	75	1.028	1.106
PLTR	Douglas	75	1.989	2.062
PLTR	Craig-Sneyd	75	1.989	2.062
PLTR	Modified Craig-Sneyd	75	1.992	2.065
PLTR	Hundsdorfer-Verwer	75	1.983	2.056
PLTR	Monte Carlo	75	1.899	1.969

Heston results were driven much more by calibration quality across tickers than by the specific ADI scheme, with all four methods giving very similar errors.

Convergence Results Heston

Scheme	Scenario	Final Abs. Error	Final Runtime (s)	Assessment
Douglas	Spatial	0.8315	2.8669	Clear convergence
Craig-Sneyd	Spatial	0.8268	5.1725	Clear convergence
Modified Craig-Sneyd	Spatial	0.5759	6.0602	Clear convergence
Hundsdorfer-Verwer	Spatial	1.2618	5.7016	Clear convergence
Douglas	Temporal	0.7886	15.5314	Clear convergence
Craig-Sneyd	Temporal	0.7850	28.3247	Clear convergence
Modified Craig-Sneyd	Temporal	0.5448	33.7469	Clear convergence
Hundsdorfer-Verwer	Temporal	1.2012	31.1534	Clear convergence

Spatial and Temporal Refinement
Conditions: $S_0 = K = 100$, $T = 1$, reference price = 10.4033

Stress Test Results Heston

Scheme	Scenario	Final Abs. Error	Final Runtime (s)	Assessment
Douglas	Stress	1.0742	1.8987	Stable, reduced accuracy
Craig-Sneyd	Stress	1.0722	3.4360	Stable, reduced accuracy
Modified Craig-Sneyd	Stress	0.9264	4.0698	Stable, reduced accuracy
Hundsdorfer-Verwer	Stress	1.3229	3.7635	Stable, reduced accuracy

Stress Comparison

Conditions: $\rho = -0.90$, $\xi = 0.60$, $S_0 = K = 100$, $T = 1$, reference price = 8.43697

Results Merton Jump-Diffusion

Ticker	Scheme	Contracts	MAE	RMSE
SPX	IMEX Euler	75	3.063	4.211
SPX	Monte Carlo	75	0.746	0.989
RUT	IMEX Euler	75	1.587	2.153
RUT	Monte Carlo	75	0.827	1.239
TSLA	IMEX Euler	75	0.321	0.447
TSLA	Monte Carlo	75	0.150	0.186
PLTR	IMEX Euler	75	0.173	0.214
PLTR	Monte Carlo	75	0.076	0.084

The results suggest jumps added limited value on much of this sample, especially where the calibrated jump parameters were small or zero.

Conclusion

Conclusion

Black-Scholes: all three PDE schemes gave nearly identical errors, with Monte Carlo often performing better.

Heston: ADI schemes behaved similarly, and calibration quality mattered more than scheme choice.

Merton: jumps added limited value on most underlyings, though TSLA showed meaningful downside jump behavior.

Main lesson: more realistic models add flexibility, but they only improve results when supported by the data and paired with careful calibration and validation.

Thank You

GitHub Repository: link